MIL-C-46501C (MU)
29 July 1965
SUPERSEDING
MIL-C-46501B (MU)
18 June 1964

MILITARY SPECIFICATION

CHARGE, DEMOLITION, LINEAR, M96 AND M58 LOADING AND ASSEMBLING

1. SCOPE

1.1 This specification covers quality assurance provisions and special requirements, not covered by the drawings, pertaining to the loading and assembling of one type of mine clearing charge designated as Charge, Demolition, Linear, M96 and M58 (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this purchase description to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-R-11470 - Radiographic Inspection, Qualification of Equipment, Operators and Procedures.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1375

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DRAWINGS

ORDNANCE CORPS

8845237 - Linear Demolition Charge, M96 and M58, Loading Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-8845237 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies shall comply with all requirements specified on Drawing (dwg.) 8845237 and with all requirements specified in applicable specifications.

3.3 Radiographic examination of detonating cord assembly.-The relay cup of the detonating cord assembly shall be examined radiographically for the presence of a gap between the PETN charge and the detonating cord. No assembly shall contain a gap in excess of 0.03 inch.

3.3.1 Radiographic qualification.-Radiographic equipment, operators and procedures shall be qualified in accordance with Specification MIL-R-11470 prior to use in production.

3.4 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, fungus, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any

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part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Statement of findings on all composition, powder and associated ingredients, and certificates of conformance on all remaining material when such material is covered by Government specifications.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of linear demolition charge assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

a. PETN from not more than one batch or homogeneous blend of batches.
b. Composition C-4 from not more than one lot.
c. Detonating cord from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used, for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Cup, relay (see dwg. 8845220 covering a detail of dwg. 8847494).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Thickness through bottom	Gage	01001
102.	Chamfer missing	Visual	01002
Minor:	AQL 1.50 percent		
201.	Inside diameter	Gage	01003
202.	Outside diameter, maximum (max.)	Gage	01004
203.	Evidence of poor workmanship (see 3.4)	Visual	01005

4.2.2.2 Relay cup, loading assembly (see dwg. 8847494 covering a detail of dwg. 8845326).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Height of charge	Gage	02001
102.	Charge missing or cracked	Visual	02002
Minor:	None defined		

4.2.2.3 Detonating cord (see dwg. 8847495 and 8847496 covering a detail of dwg. 8845236).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	End not varnished	Visual	03001
102.	Cord damaged	Visual	03002
Minor:	AQL 1.00 percent		
201.	End not square	Visual	03003
202.	Length (as applicable)	Gage	03004

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4.2.2.4 Thimble (see dwg. 8827720 covering a detail of dwg. 8845236).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Diameter of hole, minimum (min.)	Gage	04001
202.	Outside diameter, max	Gage	04002
203.	Thickness, max	Gage	04003
204.	Radius undersize	Gage	04004
205.	Protective coating missing or inadequate	Visual	04005
206.	Evidence of poor workmanship (see 3.4)	Visual	04006

4.2.2.5 Core assembly (see dwg. 8845257 covering a detail of dwg. 8845236).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Improper or insufficient splice	Visual	05001
102.	Rope damaged	Visual	05002
Minor:	AQL 1.50 percent		
201.	Length (as applicable)	Gage	05003
202.	Splice not tapered	Visual	05004
203.	Evidence of poor workmanship (see 3.4)	Visual	05005

4.2.2.6 Pellet assembly, prior to covering (see dwg. 8845227 covering a detail of dwg. 8845236).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Weight of pellet	Balance	06001
Minor: None defined			

4.2.2.7 Pellet assembly (see dwg. 8845227 covering a detail of dwg. 8845236).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Length, max	Gage	07001
202.	Improperly wrapped or sealed (when applicable)	Visual	07002
203.	Evidence of poor workmanship (see 3.4)	Visual	07003

4.2.2.8 Charge assembly, prior to assembly of charges (see dwg. 8845236 covering a detail of dwg. 8845237).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Distance between thimble centers (as applicable)	Gage	08001
102.	Detonating cords not properly assembled	Visual	08002
103.	Detonating cord missing or damaged	Visual	08003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	08004

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4.2.2.9 Charge assembly, prior to assembly of sleeves (see dwg. 8845236 covering a detail of dwg. 8845237).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Charge bag improperly folded	Visual	09001
102.	Charge missing	Visual	09002
103.	Charge improperly taped	Visual	09003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	09004

4.2.2.10 Charge assembly, (see dwg. 8845236 covering a detail of dwg. 8845237).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Sleeve missing or damaged	Visual	10001
102.	Cord improperly or loosely tied	Visual-Manual	10002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	10003

4.2.2.11 Linear charge, prior to overlapping sleeves (see dwg. 8845237).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Detonating cords not paired.....	Visual	11001
102.	Cord pairs not secured.....	Visual	11002
103.	Link nut loose.....	Manual	11003

Minor: None defined

4.2.2.12 Linear charge (see dwg. 8845237).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Adjacent sleeves not properly assembled.....	Visual	12001
102.	Cord improperly or loosely tied.....	Visual	12002
103.	Cord missing.....	Visual	12003
Minor:	AQL 1.50 percent		
201.	Detonating cords not tucked in at forward end.....	Visual	12004
202.	Sleeve covers thimble hole at forward end.....	Visual	12005
203.	Marking missing, misleading or unidentifiable.....	Visual	12006

4.2.3 Testing

4.2.3.1 Radiographic examination of detonating cord assembly (see 3.3) - Major defect, Code No. 03005.-Both relays of 75 percent of the units of each lot shall be radiographically examined. In the event one gap is discovered which exceeds the permitted size, the lot shall be examined 100 percent. Each assembly containing a gap of excessive size shall be classed defective and removed from the lot. Testing shall be as specified in 4.3.1.

4.2.3.2 Relay cup pull test (see dwg. 8845222) - Major defect, Code No. 13001.-This test shall be performed 100 percent. Any assembly which exhibits any slippage whatsoever shall be classed defective and removed from the lot. Testing shall be as specified in 4.3.2.

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4.2.3.3 Specific gravity of demolition pellet (see dwg. 8845227) - Major defect, Code No. 06002.-A sample of one pellet shall be selected approximately every two hours during production. In the event the sample fails to meet the requirement, that portion of the production represented by the sample shall be rejected. Testing shall be as specified in 4.3.3. (Sample from each extruding machine).

4.2.3.4 Die requalification.-At the conclusion of every 30, 8 hour working days (one shift) or completion of every 80,000 cycles of the die casting or extruding machine, whichever comes first, the contractor shall submit to a Government approved facility designated by the Contracting Officer each die being used in the production process in fulfilling the contract and the specifications and requirements as contained in drawing 8845236, and whenever a change occurs in the manufacturing process. Each die shall be identified by its production process number and shall be checked for the location of the groove on the die as specified in the applicable drawing. If the dies submitted are defective, thus disqualifying the dies, the contractor may correct his dies and resubmit for acceptance. Only those parts produced by qualified dies shall be acceptable for production. Die requalification is considered a special test and parts expended shall not be considered as deliveries.

4.2.3.5 Moisture content of PETN (see dwg. 8847494) - Major defect, Code No. 02003.-At the loading plant, before loading, a sample of 200 grams shall be selected for this test. In the event the sample contains moisture in excess of that permitted by the drawing, the material represented by the sample shall be rejected. Testing shall be as specified in 4.3.4.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-8845237 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation,

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production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 Radiographic examination of detonating cord assembly.-This examination shall be performed using the equipment, operators and procedures qualified under 3.3.1 with the cord assembly so positioned relative to the beam that a gap cannot be obscured by adjacent material. Each assembly shall be identified to its position on the radiograph and the exposure made. Observation shall be made for the presence of gaps of excessive size. Any assembly containing such a defect shall be classed defective.

4.3.2 Relay cup pull test.-This test shall be performed using the quipment specified in 4.2.4. Prior to performance of the test, ink of a color contrasting to that of the detonating cord shall be applied to the end of the relay cup and to 1/8 inch, minimum, of the detonating cord a full 360 degrees. After completion of the test, observation shall be made for withdrawal of the cord from the cup. Any assembly exhibiting slippage shall be classed defective.

4.3.3 Specific gravity of demolition pellet.-The pellet shall be weighed on a balance and the weight recorded as dry weight. It shall then be submerged under water in such a manner

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that there is no entrapped air and then weighed. The temperature of the water shall be maintained at 68 degrees Fahrenheit (F.) plus or minus 2 degrees F. The second weight shall be recorded as weight under water. Specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{A}{A-B}$$

where:
A = dry weight.
B = weight under water.

4.3.4 Moisture content of PETN.-An accurately weighed portion of approximately 40 grams of the sample shall be transferred to a tared weighing dish. The dish and contents shall be placed in an oven, which shall be maintained at 100 degrees Centigrade (C.) plus or minus 2 degrees C., for 2 hours and then weighed. Determine the loss in weight and calculate the percentage of moisture in the portion.

5. PREPARATION FOR DELIVERY

5.1 Interim storage.-Pending assembly into Demolition Kit, Projected Charge, M173 or M125, the linear charges shall be coiled on pallets or in boxes and shall be stored in a dry place.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:
a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 -Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

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6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35MM microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set.

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6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Intended use.-It is intended that the item of this specification will be used in Demolition Kit, Projected Charge, M173 and Charge, Demolition, M58.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A947

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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